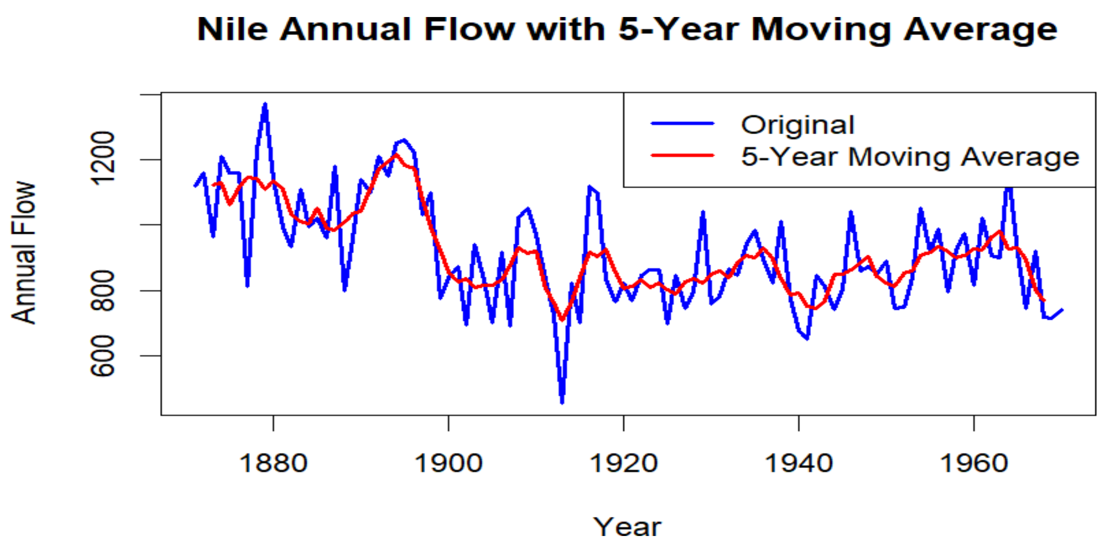
ASSESSMENT TOOL-2
Visualization Evaluation Exercise

Code of Conduct:

*I **C Nithisha (192424375)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C Nithisha

Question 1: Individual Time Series & Moving-Average Smoothing



Code:

```
nile_ts <- ts(Nile, start = 1871, frequency = 1)
ma5 <- filter(nile_ts, rep(1/5, 5), sides = 2
plot(nile_ts,
    type = "l",
    col = "blue",
    lwd = 2,
    xlab = "Year",
    ylab = "Annual Flow",
    main = "Nile Annual Flow with 5-Year Moving Average")
```

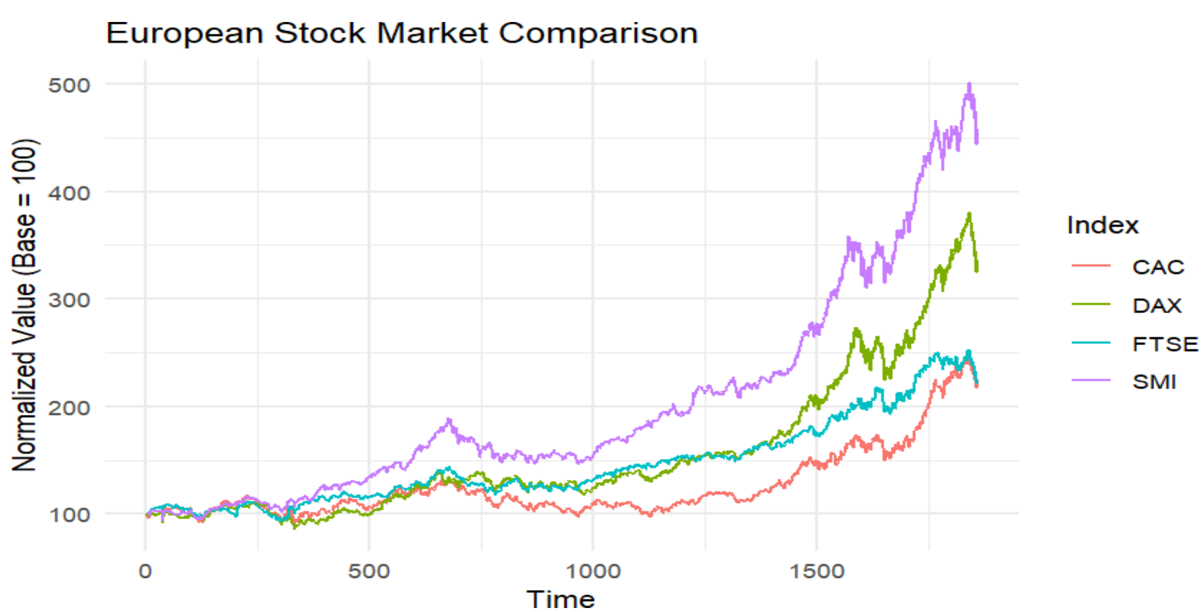
```
lines(ma5, col = "red", lwd = 2)
legend("topright",
       legend = c("Original", "5-Year Moving Average"),
       col = c("blue", "red"),
       lty = 1,
       lwd = 2)
```

Observation

The Nile series shows a noticeable change in level around the late 1890s to early 1900s, where the annual flow appears to decrease. The moving-average line makes this shift clearer by reducing short-term fluctuations. The variability also appears different before and after the level change.

Question 2: Multiple Time Series Comparison & Dose–Response Curve

Part A: EuStockMarkets Dataset



Code:

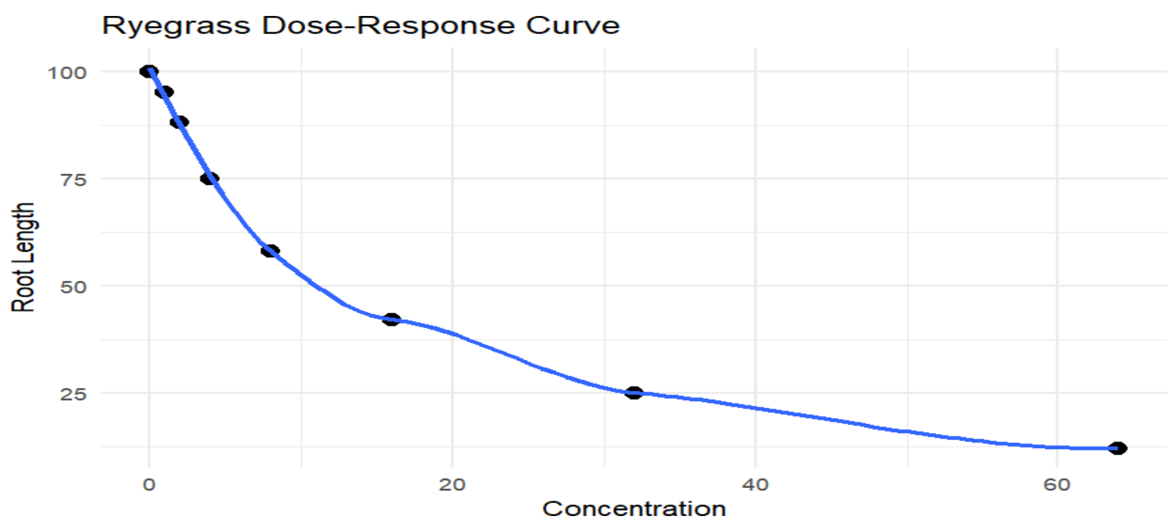
```
library(ggplot2)
```

```
library(tidyr)

library(dplyr)

stock_data <- as.data.frame(EuStockMarkets)

stock_data$Time <- 1:nrow(stock_data)

stock_long <- stock_data %>%

  pivot_longer(

    cols = c(DAX, SMI, CAC, FTSE),

    names_to = "Index",

    values_to = "Value")

stock_normalized <- stock_long %>%

  group_by(Index) %>%

  mutate(Normalized_Value = Value / first(Value) * 100)

ggplot(stock_normalized,

       aes(x = Time,

           y = Normalized_Value,

           color = Index)) +

  geom_line() +

  labs(

    title = "European Stock Market Comparison",

    x = "Time",

    y = "Normalized Value (Base = 100)"  ) +

  theme_minimal()
```

Observation

The index whose normalized line finishes at the highest value shows the strongest overall growth during the period. After re-indexing all series to 100, their relative percentage growth can be compared even though their original values are on different scales

Part B: Ryegrass Dose-Response



Code:

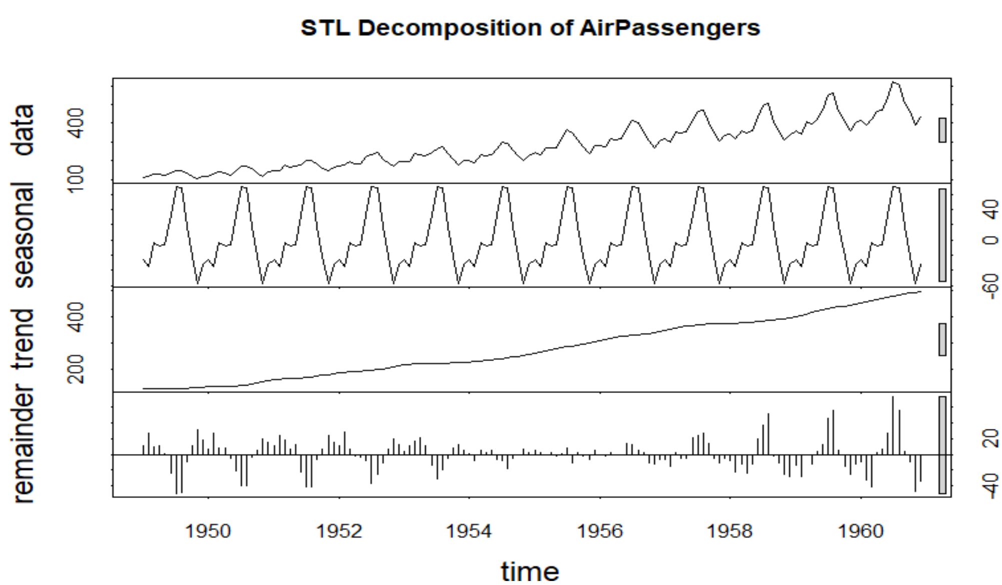
```
library(ggplot2)

ryegrass <- data.frame(

  concentration = c(0, 1, 2, 4, 8, 16, 32, 64),

  root_length = c(100, 95, 88, 75, 58, 42, 25, 12))

print(ryegrass)

ggplot(ryegrass,

       aes(x = concentration,

           y = root_length)) +

  geom_point(size = 3) +

  geom_smooth(method = "loess", se = FALSE) +
```

```
  labs(
    title = "Ryegrass Dose-Response Curve",
    x = "Concentration",
    y = "Root Length"  ) +
  theme_minimal()
```

Observation

As concentration increases, root length generally decreases, producing a negative dose–response relationship. The concentration returned in `result` is the approximate concentration at which root length is reduced to about 50% of the control value.

Question 3: STL Decomposition of AirPassengers



Code:

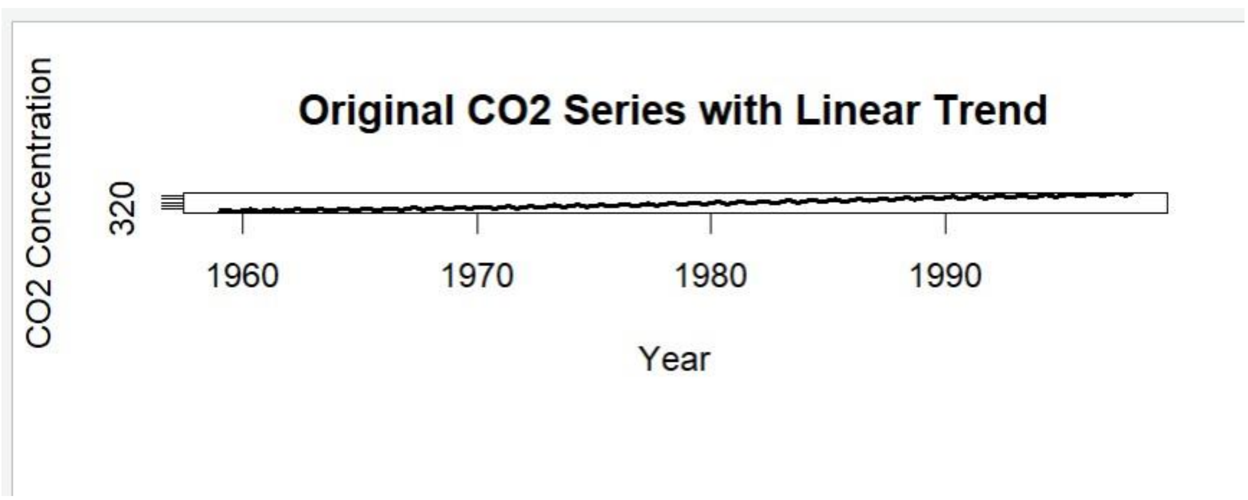
```
air_ts <- ts(
  AirPassengers,
  start = c(1949, 1),
  frequency = 12)
```

```
air_stl <- stl(
  air_ts,
  s.window = "periodic")
plot(air_stl,
     main = "STL Decomposition of AirPassengers")
```

Observation

The AirPassengers data shows an increasing trend over time along with strong yearly seasonal patterns. The seasonal fluctuations become larger as the number of passengers increases, so a **multiplicative decomposition** is more appropriate than a purely additive decomposition.

Question 4: Detrending and Residual Analysis of CO_2



Code:

```
data(co2)
co2_values <- na.omit(co2)
time_values <- time(co2_values)
co2_numeric <- as.numeric(co2_values)
model <- lm(co2_numeric ~ time_values)
trend <- fitted(model)
```

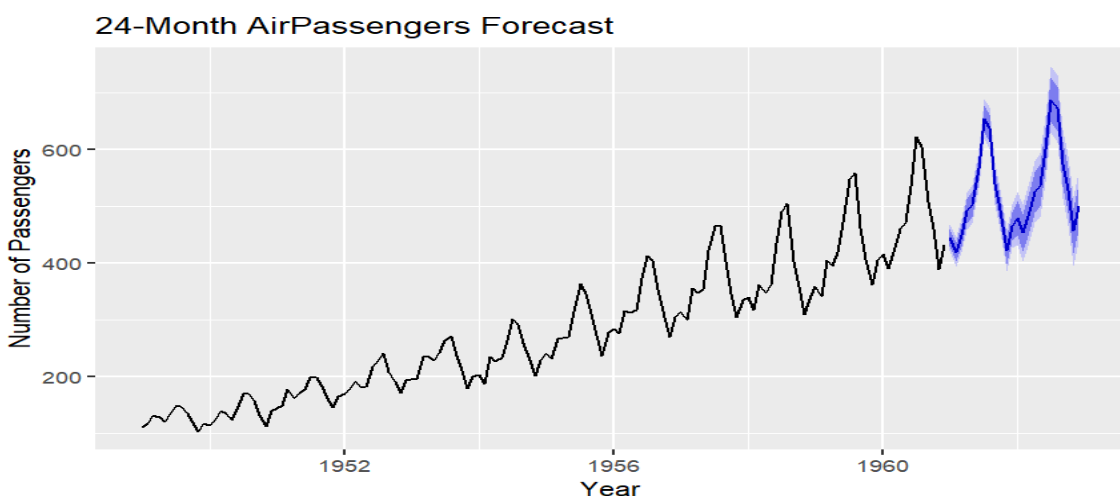
```
detrended <- residuals(model)

par(mfrow = c(2, 1))

plot(time_values,

     co2_numeric,

     type = "l",

     xlab = "Year",

     ylab = "CO2 Concentration",

     main = "Original CO2 Series with Linear Trend",

     lwd = 2)

lines(time_values,

      trend,

      lwd = 2,

      col = "red")

legend("topleft",

       legend = c("Original CO2", "Linear Trend"),

       col = c("black", "red"),

       lty = 1,

       lwd = 2)

plot(time_values,

     detrended,

     type = "l",

     xlab = "Year",

     ylab = "Residual",
```

```
    main = "Detrended CO2 Residuals",
    lwd = 2)
abline(h = 0,
      col = "red",
      lty = 2)
par(mfrow = c(1, 1))
```

Observation

After removing the linear trend, a clear repeating seasonal pattern remains in the residuals. Therefore, simple linear detrending is not sufficient because the CO_2 series contains both a long-term upward trend and strong seasonal variation.

Question 5 – Part A: Forecasting AirPassengers



CODE:

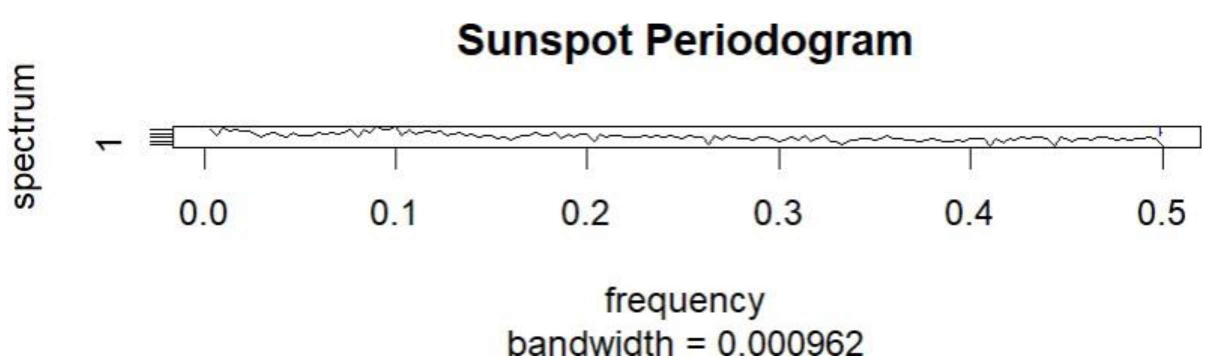
```
install.packages("forecast")
library(forecast)
air_ts <- AirPassengers
model <- auto.arima(air_ts)
```

```
forecast_result <- forecast(
  model,
  h = 24)
autoplot(forecast_result) +
  ggtitle("24-Month AirPassengers Forecast") +
  xlab("Year") +
  ylab("Number of Passengers")
```

Observation

The forecast continues the repeating yearly seasonal pattern observed in the historical AirPassengers data. The peaks and low points occur in a similar cycle, indicating that the model has captured the seasonality reasonably well.

Part B: Sunspot Cycle Detection



CODE:

```
sunspot_ts <- sunspot.year
acf(
  sunspot_ts,
  main = "ACF of Sunspot Data"
```

```
)

spectrum(
  sunspot_ts,
  main = "Sunspot Periodogram"
)
```

Observation

The ACF and periodogram typically indicate a repeating cycle of approximately **10–11 years**. This is consistent with the commonly cited solar cycle of approximately **11 years**, although the exact cycle length can vary slightly.